

Supervised, Unsupervised, and Reinforcement Learning

A plain-language companion to Session 2, slide 15

A **learning paradigm** describes how a machine-learning system receives feedback while learning. It answers this question:

What information tells the system whether it is learning something useful?

The paradigm is not the task, the data format, or the model architecture. For example, classification is a **task**, an image is a **data modality**, supervised learning is a **learning paradigm**, and a CNN is a **model class**.

The three paradigms in one minute

Paradigm	What the system receives	What it learns to do	Memory phrase
Supervised	Examples paired with known answers	Predict the answer for a new example	Learn from an answer key
Unsupervised	Examples without answer labels	Find groups, structure, or unusual cases	Find structure without an answer key
Reinforcement	Actions followed by rewards or penalties	Choose actions that earn more total reward	Learn from consequences

These are different learning arrangements, not three levels of intelligence. The correct choice depends on the problem and on what feedback is actually available.

1. Supervised learning: learn from an answer key

The simple idea

Supervised learning uses historical examples where the desired answer is already recorded. The system compares its prediction with that answer and adjusts to make future predictions less wrong.

TRAINING

```
input example + known answer
  |
  v
MODEL makes a prediction
  |
  v
prediction compared with known answer
  |
  v
model adjusts and repeats
```

AFTER TRAINING

new input -> trained model -> predicted label, number, or score

A useful analogy is studying with practice questions that include an answer key. You answer a question, check the correct answer, learn from the mistake, and try again.

Fraud-lab example

Each historical transaction has input information such as amount, merchant category, time, and card history. It also has a later outcome label:

- `is_fraud = 1`: known fraud;
- `is_fraud = 0`: not labeled as fraud.

```
transaction features + known fraud outcome
      |
      v
supervised model training
      |
      v
score a future transaction
```

The random forest in Applied AI Studio is supervised because it learns from transactions paired with known outcomes.

Common tasks

- classification: fraud or normal, defect or no defect;
- regression: predict a delivery time or dollar amount;
- forecasting when historical outcomes are available;
- medical-image classification using expert labels; and
- document routing using correctly assigned examples.

What can go wrong

- The labels may be wrong, incomplete, biased, or delayed.
- The model may learn past decisions instead of true outcomes.
- The training examples may not represent future cases.
- A label may reveal information that was unavailable at decision time, causing leakage.

In fraud detection, a chargeback may arrive months after the purchase. A transaction blocked before completion may never receive a reliable fraud outcome. Having a label column does not automatically make the label trustworthy.

Question to ask

What exactly does the label mean, who created it, and when did it become available?

2. Unsupervised learning: find structure without an answer key

The simple idea

Unsupervised learning receives examples but no correct-answer label. It searches for patterns in how the examples resemble or differ from one another.

```
unlabeled examples
  |
  v
compare similarities and differences
  |
  +--> groups or clusters
  +--> compressed patterns
  +--> unusual observations
```

A useful analogy is sorting a mixed box of objects without being told the categories. You might group them by shape, color, or size. The groups come from patterns in the objects, not from an answer key.

Examples

Customer segmentation: Group customers whose purchasing behavior is similar, even though nobody supplied labels such as "budget shopper" or "frequent traveler."

Anomaly detection: Learn what ordinary sign-in behavior looks like and flag a sign-in that is far from that pattern.

Common tasks

- clustering similar customers, documents, or products;
- finding unusual transactions or system events;
- reducing many variables to a smaller representation;
- exploring a new dataset before labels exist; and
- discovering possible patterns for expert investigation.

What can go wrong

- A mathematical cluster may have no useful business meaning.
- An unusual case is not automatically a bad case.
- Results can change when features, distance measures, or the number of groups change.
- Human interpretation is still needed to name and validate discovered patterns.

An anomaly detector can say, "This transaction is unusual compared with the history." It cannot conclude, "This transaction is fraud," unless later evidence supports that conclusion.

Question to ask

How will a domain expert decide whether the discovered groups or anomalies are meaningful and useful?

3. Reinforcement learning: learn from consequences

The simple idea

In reinforcement learning, an **agent** takes an action in an environment. The environment returns a reward or penalty and a new situation. Over many attempts, the agent learns a **policy**, meaning a strategy for choosing actions that produces more total reward.



A useful analogy is learning a game through repeated play. Winning provides positive feedback, losing provides negative feedback, and the player gradually improves the strategy.

Example

A warehouse robot must learn how to move an item without collisions. It receives positive reward for completing the task and penalties for collisions, wasted movement, or excessive time. The important question is not whether one action was labeled "correct." It is whether a sequence of actions produced a good result.

Common tasks

- game-playing systems;
- robotics and control;
- selected resource-allocation problems;
- simulated navigation; and
- sequential decisions where one action changes what happens next.

What can go wrong

- A poorly designed reward can teach the wrong behavior.
- The agent may find a shortcut that earns points without achieving the real goal.
- Learning through real-world trial and error can be costly or unsafe.

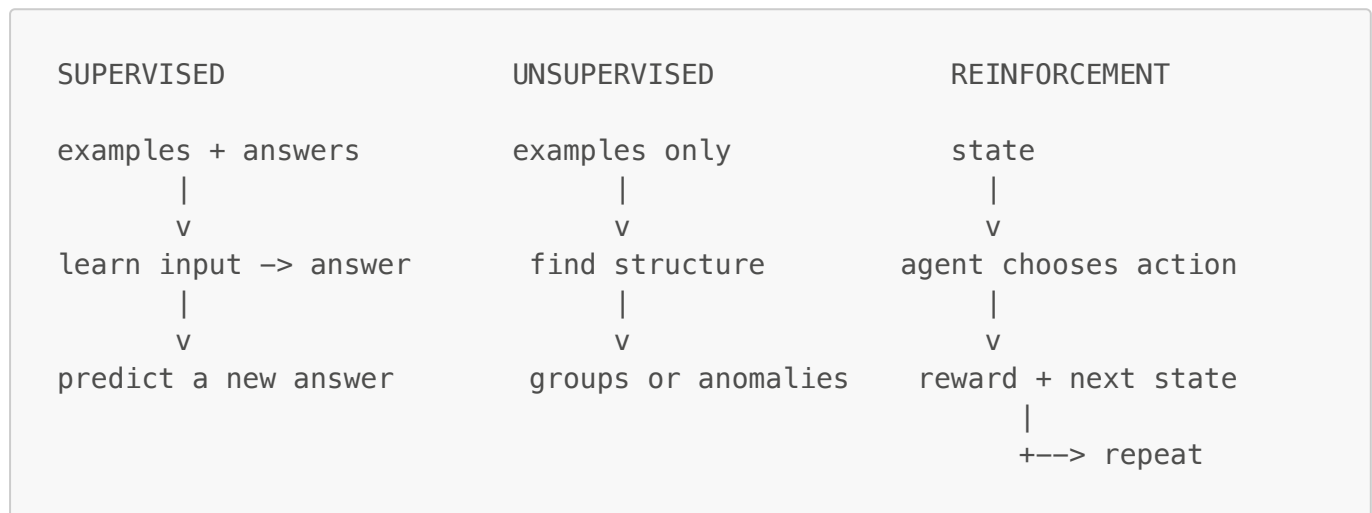
- A reward may arrive long after the action that caused it.
- Exploration is difficult when a bad action could harm a person or business.

For this reason, reinforcement learning is often developed in games or simulations before any real-world use. It is uncommon in routine business-analysis projects.

Question to ask

What behavior does the reward actually encourage, and is exploration safe?

Side-by-side visual



The most important question: do we have labels?

Do historical examples have trustworthy answers?

YES

- > Supervised learning may be possible.
- > Ask what the labels mean and when they became available.

NO

- > Do we want to discover groups or unusual cases?
YES -> Consider unsupervised learning.
- > Is this a sequence of actions with measurable consequences?
YES -> Reinforcement learning may be relevant.
- > Is the decision already fully defined by policy or arithmetic?
YES -> Use rules or optimization; machine learning may not be needed.

Do not confuse task with learning paradigm

The **task** says what result must come out. The **learning paradigm** says how the system learns to produce it.

Question	Dimension	Example answer
What should come out?	Task	Fraud-risk classification
What kind of data goes in?	Data modality	Tabular transaction data
How does it learn?	Learning paradigm	Supervised learning
What is built?	Model class	Random forest

One task can sometimes be approached through different paradigms:

- Known fraud labels available: supervised classification.
- No labels, but unusual behavior matters: unsupervised anomaly detection.
- Repeated actions change future outcomes: possibly reinforcement learning.

The outputs are not equivalent. "Unusual" is not the same claim as "fraud."

Important extension: self-supervised learning

Slide 15 emphasizes the three paradigms students will encounter most often at this level. Elsewhere in the deck, **self-supervised learning** appears because it is central to modern language models.

The simple idea

Self-supervised learning starts with data that people did not manually label. The training process creates an answer from the data itself, often by hiding or predicting part of it.

```
original text: The customer paid the invoice.  
  
training input: The customer [MASK] the invoice.  
answer created from the text: paid  
                        |  
                        v  
model learns language patterns
```

Another common objective is predicting the next token from the earlier tokens. This lets a model learn from vast amounts of text without humans labeling every sentence.

Self-supervised does not mean the machine supervises itself like a person. It means the training target is derived automatically from the data.

Large language models are generally pretrained this way and then may be further tuned with human-created examples, preference feedback, and other methods.

Question to ask

What was the model trained to predict, and how does that training objective differ from the business task we want it to perform?

Examples across the course

Scenario	Likely paradigm	Why
Learn fraud risk from transactions with chargeback outcomes	Supervised	Each historical example has an outcome label
Group customers by similar purchasing behavior	Unsupervised	The groups are not supplied in advance
Flag sign-ins unlike normal behavior	Unsupervised	The system models normal patterns without needing every attack label
Predict appointment no-shows from known attendance outcomes	Supervised	Past appointments have attended or missed outcomes
Train a robot to navigate using rewards and penalties	Reinforcement	Actions affect later states and rewards
Pretrain a language model by predicting hidden or next words	Self-supervised	The text generates its own training target
Reject a form missing required fields	Neither; use rules	The correct logic is already explicit and stable

What students should remember

1. **Supervised:** Learn from examples with known answers.
2. **Unsupervised:** Find structure in examples without known answers.
3. **Reinforcement:** Learn a strategy from rewards produced by actions.
4. **Self-supervised:** Create training targets from the data itself, often for pretraining.
5. A label must be trustworthy, not merely present.
6. An anomaly is unusual, not automatically wrong or fraudulent.
7. A reward defines what an agent will pursue, so reward design is a business and safety decision.
8. Not every problem needs machine learning.

Five-minute class activity (Optional)

For each scenario, choose **supervised**, **unsupervised**, **reinforcement**, **self-supervised**, or **not machine learning**. Explain which learning feedback is available.

1. Predict whether a payment is fraud using transactions with confirmed outcomes.
2. Group stores by purchasing patterns when no store types have been defined.
3. Train a simulated delivery robot using points for safe, timely completion.
4. Learn general language patterns by predicting missing words in millions of documents.
5. Reject an application when a required signature is absent.
6. Flag a network login that is very different from ordinary employee behavior.

Suggested answers

1. **Supervised:** Each historical transaction is paired with a known fraud outcome.
2. **Unsupervised:** The system discovers groups without predefined group labels.

3. **Reinforcement:** Actions receive rewards and change the robot's next situation.
4. **Self-supervised:** The documents automatically provide the missing-word answers.
5. **Not machine learning:** A deterministic required-field rule is sufficient.
6. **Unsupervised:** The system identifies deviation from the learned normal pattern; unusual still requires investigation.